MIL-T-47052(MI)
10 May 1974
SUPERSEDING
MIS-13874
11 September 1967

MILITARY SPECIFICATION

TAPE, EPOXY-GLASS CLOTH, THERMOSETTING, PRESSURE SENSITIVE ADHESIVE

This specification is approved for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. The specification covers one type of epoxy-glass cloth, thermosetting, pressure sensitive adhesive tape.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents, of the issue in effect on the date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATION

Military

MIL-P-14232	Parts, Equipment and Tools for Army Materiel, Packaging and Packing of
MIL-I-15126	Insulation Tape, Electrical Pressure Sensitive Adhesive and Pressure Sensitive Thermosetting Adhesive

FSC 5970

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STANDARD

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD 129	Marking for Shipment and Storage

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of the specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials

ASTM D 149	Dielectric Breakdown Voltage and Dielectric Strength of Electrical Insulation Materials at Commercial Power Frequencies, Tests for,
ASTM D 257	D-C Resistance or Conductance of Insulating Materials, Tests for,

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

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3. REQUIREMENTS

3.1 Preproduction sample. Unless otherwise specified (see 6.2), a preproduction sample shall meet all the requirements of this specification.

3.2 Materials.

3.2.1 Composition. The material shall be an electrical adhesive tape having an epoxy impregnated glass fabric backing and a pressure sensitive thermosetting adhesive.

3.2.2 Dimensions.

3.2.2.1 Width. The tape shall be furnished in the standard widths and tolerances as specified for Type GMT of MIL-I-15126.

3.2.2.2 Thickness. The thickness shall be 0.005 plus or minus 0.0001 inch.

3.2.3 Roll size. The dimensions for roll diameter and length shall be as specified for Type GFT of MIL-I-15126.

3.3 Performance characteristics.

3.3.1 Tensile strength. The tape shall have a minimum average tensile strength of 80 pounds per inch width.

3.3.2 Adhesion. The tape shall have a minimum average adhesion to steel of 50 ounces per inch width.

3.3.3 Indirect electrolytic corrosion. The tape shall have an indirect electrolytic corrosion as indicated by electrical conductance, of not more than 100 picomho (10^{-10} mho).

3.3.4 Dielectric strength. The tape shall not break down when subjected to not less than 4000 volts root mean square (rms).

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3.3.5 Insulation resistance. The tape shall have a minimum average insulation resistance of 1 x 10^9 ohms.

3.4 Environmental conditions.

3.4.1 Thermosetting. The tape shall have an adhesion backing, in ounces per inch width, of 40 minimum before and 100 minimum after 2 hours at 130 plus or minus 2 degrees Celsius (C) (266 plus or minus 3.6 degrees Fahrenheit (F)).

3.5 Workmanship. The adhesive surface shall be uniform and smooth, without bare spots or lumps. The rolls shall be uniformly and symmetrially wound. There shall be no visible overlap of the edges of the tape or separation between layers.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction inspection. Unless otherwise specified in the contractual document (see 6.2), a sample consisting of one roll containing 25 feet of tape, manufactured under the same conditions as those proposed for subsequent production, shall be subjected to preproduction inspection. Accomplishment of preproduction inspection shall be as specified in the contract. Subsequent units will not be considered for acceptance until Government approval of the preproduction sample has been obtained. Units subjected to preproduction inspection shall have successfully passed acceptance inspection. Testing of the preproduction sample, to determine compliance with the

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characteristics listed in Table I, shall be conducted in accordance with the corresponding test paragraphs.

Table I

Preproduction inspection		
Characteristic	Requirement Source	Test Paragraph
Tensile strength	3.3.1	4.5.3.4
Indirect electrolytic corrosion	3.3.3	4.5.3.6
Dielectric strength	3.3.4	4.5.3.7
Thermosetting	3.4.1	4.5.3.9

4.3 Quality conformance inspection.

4.3.1 Sampling for acceptance inspection. When sampling for acceptance inspection is specified, it shall be conducted in accordance with MIL-STD-105, at an acceptable quality level (AQL) of 4.0 percent defective.

4.3.2 Lot formation. A lot shall consist of all the material manufactured in one continuous operation by the same process, by the same manufacturer in accordance with this specification and submitted for inspection at one time.

4.3.3 Acceptance inspection. Acceptance inspection of the sample specified in 4.3.1, to determine compliance with the characteristics specified in Table II, shall be conducted in accordance with the corresponding test paragraphs.

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4.4 Inspection equipment. The inspection equipment for conducting examination and tests shall be as specified in the applicable test methods and procedures paragraphs.

Table II

Acceptance inspection		
Characteristic	Requirement Source	Test Paragraph
Composition	3.2.1	4.5.3.1
Dimensions	3.2.2	4.5.3.2
Roll size	3.2.3	4.5.3.3
Adhesion	3.3.2	4.5.3.6
Insulation resistance	3.3.5	4.5.3.8
Workmanship	3.5	4.5.5
Packaging, packing and marking	5.1	4.5.4

4.5 Test methods and procedures.

4.5.1 Test conditions. Unless otherwise specified herein, the following conditions shall be used as a basis to establish performance requirements.

a. Temperature, room ambient (plus 25 plus or minus 5 degrees Celsius (C) (plus 77 plus or minus 9 degrees Fahrenheit (F)).

b. Altitude, facility ground.

c. Humidity, facility ambient up to 95 percent relative humidity.

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4.5.2 Test sequence. Testing within each classification of inspection (preproduction or acceptance) shall be at the option of the contractor.

4.5.3 Tests.

4.5.3.1 Composition. Composition shall be examined visually to establish compliance to 3.2.1.

4.5.3.2 Dimensions. The tape shall be measured in accordance with MIL-I-15126. The tape shall conform to the requirements of 3.2.2.

4.5.3.3 Roll size. Roll dimensions shall be measured as specified in MIL-I-15126, Type GFT, and shall conform to 3.2.3.

4.5.3.4 Tensile strength. Testing for tensile strength shall be as specified in MIL-I-15126, to establish conformance with 3.3.1.

4.5.3.5 Adhesion. Testing for adhesion shall be as specified in MIL-I-15126, to establish conformance with 3.3.2.

4.5.3.6 Indirect electrolytic corrosion. The test for indirect electrolytic corrosion shall be conducted as specified in MIL-I-15126, to establish conformance with 3.3.3.

4.5.3.7 Dielectric strength. Testing for dielectric strength shall be in accordance with ASTM D 149, to establish conformance with 3.3.4.

4.5.3.8 Insulation resistance. Testing for insulation resistance shall be in accordance with ASTM D 257, to establish conformance with 3.3.5.

4.5.3.9 Thermosetting. Testing for thermosetting shall be in accordance with MIL-I-15126. The tape shall meet the requirements of 3.4.1.

4.5.4 Packaging, packing and marking. The inspector shall ascertain that the packaging, packing and marking of the material conforms to this specification.

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4.5.5 Workmanship. The tape shall be visually examined to assure conformance to the workmanship requirement specified in 3.5.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing and marking. Unless otherwise specified by the contract or purchase order, preservation, packaging, packing, and marking requirements for the specified levels of protection shall be in accordance with MIL-P-14232.

6. NOTES

6.1 Intended use. The tape, in accordance with this specification, is intended for use in guided missile application.

6.2 Ordering data. Procurement documents shall specify the following:

a. Title, number, and date of this specification.

b. Whether a preproduction sample is required (see 3.1), and if so, pertinent details.

c. Size required and quantity.

6.3 Supersession data. This specification includes the requirements of Missile Interim Specification MIS 13874, dated 11 September 1967.

Custodian:
Army-MI

Preparing Activity:
Army-MI
Project No. 5970-A459

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